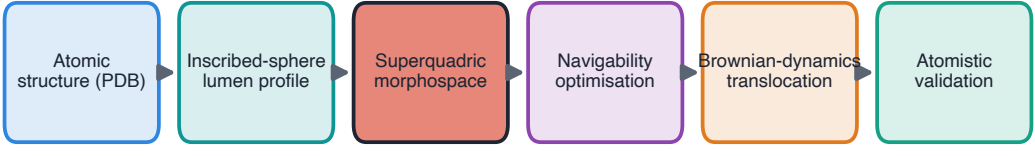Computational pipeline